

Logistic Regression, Thresholds, and the Module 4 Deliverable

Module 4, Part 3 (November 13)

QBA 5330

From Numbers to Yes/No

- Parts 1–2: predicting a **continuous** outcome (sales, revenue)
- Now: predicting a **binary** outcome (churn: yes or no)

Why Not Just Use Linear Regression?

```
1 library(tidyverse)
2 set.seed(2025)
3 n <- 300
4 tenure <- round(runif(n, 1, 60))
5 logodds_true <- 1.5 - 0.08 * tenure
6 churned <- rbinom(n, 1, 1 / (1 + exp(-logodds_true)))
7 customers <- tibble(tenure = tenure, churned = churned)
8
9 predict(lm(churned ~ tenure, data = customers), newdata = data.frame(tenure = 100))
1
-0.6852383
```

A “probability” of about -0.69 . Not usable.

The Churn Model, Live

```
1 model <- glm(churned ~ tenure, data = customers, family = binomial)
2 coef(model)
```

```
(Intercept)      tenure
1.71987908 -0.08458567
```

```
1 exp(coef(model)["tenure"])
```

```
tenure
0.9188929
```

- Odds ratio ≈ 0.919 : each month of tenure, about 8.1% lower odds of churn

One Customer, One Prediction

```
1 predict(model, newdata = data.frame(tenure = 12), type = "response")
```

1

0.6692624

- A 12-month customer: about a 67% chance of churning

Activity: Choose Your Threshold (25 minutes)

- Retention offer: **\$50** per customer sent
- Value of a retained customer: **\$600**
- Threshold 0.5: sensitivity 63.2%, specificity 85.1%
- Threshold 0.3: sensitivity 84.9%, specificity 69.1%
- **Which threshold do you recommend, and why?**

The Numbers to Work With

Threshold	Flagged	Cost	Caught	Value	Net
0.5	96	\$4,800	67	\$40,200	?
0.3	150	\$7,500	90	\$54,000	?

Debrief

- With *these* costs, the lower threshold wins
- What would have to be true for the higher threshold to win instead?

The Module 4 Deliverable

- Build a **predictive model** (regression or logistic regression) on a business dataset
- **Diagnose** it
- Translate the result into a **specific managerial recommendation**
- Calculated in R and Excel

Looking Ahead to December 4

Module 4 added build a model to the decision cycle.

Next: **forecasting**, predicting a value forward in **time**, culminating in the Ceres Gardening Company case.